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Lead-Time-Scaled Chance Constraints With a Cost-Aware Fallback for Fast Multi-UAV Replanning Under Uncertain Execution

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ABSTRACT Multi-UAV replanners usually plan for nominal execution, so under wind, speed and service-time errors, tasks near the end of a route are served late. On development flights, the spread of the finish-time error grew linearly with the lead time, because disturbances persist over a leg. The proposed replanner, LTCC, exploits this structure. With an affine error model, a chance constraint on the finish time of every task reduces exactly to an effective deadline, which fast heuristics reuse unchanged. A cost-aware fallback attempts a task that the constraint gives up only if its probability of meeting the deadline outweighs the cost of a late arrival, again in closed form and without tuning. In seven studies on held-out missions of a kinematic simulator, each with a protocol frozen in advance, the chance-constrained model served more reward than the nominal model and a constant time buffer on three independent sets of seeds, by 5.07 and 4.22 percentage points on the last one, and 10.0% more than the consensus-based bundle algorithm on a further set. A tuned proportional buffer and moment propagation came within 0.38 points. When a late arrival cost one task reward, the cost-aware fallback outscored the nominal fallback by 2.42 points and a proportional buffer tuned for this cost by 0.90 points (both $p < 0.001$), and it improved moment propagation as well. A rule matched to the late-cost model of the score beat the tuned buffer again on new seeds, by 1.34 points. At twice this cost, the rule showed no significant advantage over the tuned buffer, which remained better when the late cost was proportional to the task reward. Under four tested departures from the error model, namely heavy tails, rare long delays, a speed bias per vehicle and heavy-tailed latency, the advantages persisted. In PX4 software-in-the-loop flights, the fallback raised the utility by 56%, whereas the advantage over the nominal model was not confirmed.

INDEX TERMS Anytime planning, chance-constrained planning, dynamic vehicle routing, execution uncertainty, multi-UAV task allocation, PX4 software-in-the-loop.

I. INTRODUCTION

Teams of unmanned aerial vehicles (UAVs) are used for inspection, delivery and monitoring. During a mission, new tasks may appear, a vehicle may fail, or a deadline may be moved forward. A central planner then repairs the routes of the whole fleet within a fraction of a second, while some UAVs may already wait for new instructions. Dynamic vehicle routing and multi-robot task allocation offer many methods for this repair [1]–[4], and auction methods such as the consensus-based bundle algorithm (CBBA) [5] or metaheuristics such as adaptive large neighbourhood search (ALNS) [6] find good plans within such budgets.

Most of these replanners plan for nominal execution. When UAVs meet wind, fly slower than commanded or take longer at a task, part of the slack that the plan promises is consumed by errors that the planner does not see, and tasks near the end of a route are served late. The error grows with the time until a task is reached, so a constant time margin is too large for early tasks and too small for late ones. Stochastic and robust routing protect schedules against such errors [7]–[10], and chance-constrained allocation bounds the probability of failure [11], [12]. These models are usually solved by sampling or by exact methods that are too slow for the time available between two events. A second line of work treats the computation time of an anytime planner as a decision [13]–[15]. It is not known which of the two, the planning model or the control of computation, matters more for a replanner that faces uncertain execution.

A chance constraint raises a further question. It gives up every task whose probability of being served on time falls

below its target, even when a vehicle has nothing better to do, and whether such a task is still worth attempting depends on what a late arrival costs. Soft time windows price lateness in the objective [16], but they need the distribution of the lateness of every route inside the search. A margin without a probability, such as a constant or proportional buffer, cannot answer the question and has to be tuned again whenever the cost changes.

This paper addresses four research questions. RQ1: how does the finish-time error of a replanned UAV fleet grow along a route, and can a chance constraint built on it run inside fast heuristics? RQ2: which of the planning model, the stopping rule and the trigger matters most? RQ3: how does the resulting replanner compare with baselines from the literature, including time buffers, moment propagation and a published task-allocation algorithm, and does it carry over to PX4 autopilots? RQ4: when a late arrival has a cost, can the probability of meeting the deadline decide which given-up tasks to attempt, without tuning?

To answer them, the paper proposes LTCC (lead-time-scaled chance constraints), a replanner that plans against a chance constraint on the finish time of every task and keeps the fast heuristics of the nominal model. Its central observation is that the probability of meeting a deadline, which a buffer lacks, lets a replanner price lateness without tuning. Neither Gaussian chance constraints, nor feasibility checks under uncertainty that plug into an existing routing heuristic [8], nor ALNS or CBBA are new. The contribution lies in the interface between them and in the decision rule built on it, and it is threefold. The first is the planning model. Because wind and airspeed errors persist over a leg, the spread of the finish-time error, calibrated from closed-loop execution, grows linearly with the lead time and not with its square root, as independent increments would imply. With an affine error model, a chance constraint on every task reduces exactly to an effective deadline, which the repair and the search of the nominal model use unchanged (Proposition 1). A cost-aware fallback attempts a given-up task only if its probability of meeting the deadline outweighs the cost of a late arrival. In closed form, this rule is again an effective deadline per task, for a late cost proportional to the reward as well as for a fixed one, and it has no parameter to tune (Proposition 2). The rule needs only a probability of meeting the deadline, so it also applies to moment propagation. The second is a sequence of seven studies on held-out missions, each with a protocol frozen before its held-out runs, with re-implemented baselines. These include stopping rules from the metareasoning literature [13], [14], [17], fixed and size-scaled budgets, a constant time buffer [7], a proportional buffer, moment propagation along the route [8], CBBA [5] and a conventional event-triggered planner. Competitors are tuned for every cost of a late arrival, ablations separate the effects of the planning model, the fallback, the stopping rule and the trigger, and a final study tests execution errors that the calibrated model does not describe. The third is a software-in-the-loop evaluation with PX4 autopilots [18] in Gazebo [19] and ROS 2 [20], after a rescaling that keeps every dimensionless ratio of the decision problem, with a number of flights computed beforehand from the kinematic tier.

The results are reported as they came out. In the kinematic tier, the chance-constrained model served more reward than the nominal model and a constant time buffer on three independent sets of held-out seeds, while the choice of stopping rule mattered little. Without fallback, it was too cautious under high disturbances with halved windows, and the fallback, designed after this result and tested on new seeds, removed the weakness. A tuned proportional buffer and moment propagation, whose margins also grow with the lead time, reached the same utility within 0.38 percentage points, so the shape of the margin and the fallback carry the gain in utility rather than the precise error model. Re-scoring these runs showed that the nominal fallback pays off only if a late arrival costs less than about one task reward. With a late arrival that costs one task reward, the cost-aware fallback served 2.42 percentage points more than the nominal fallback, and 0.90 and 1.78 points more than a proportional buffer and a safety factor tuned for this cost. At twice this cost, the tuned buffer was better by 1.02 points. On a further set of seeds, the replanner served 4.86 points, or 10.0%, more utility than CBBA. With a late cost of one task reward, CBBA itself gained 12.12 points when it planned on the proposed model, and the cost-aware fallback improved moment propagation as well. Studies 5 and 6 had run the rule for a late cost proportional to the reward but scored it with a fixed cost per late arrival. On a further set of seeds, the rule matched to the score did not differ significantly from the mismatched one at one task reward, and beat the buffer tuned for this score by 1.34 points. At twice the cost, it no longer differed significantly from the tuned buffer, against which the mismatched rule had been significantly behind. Under the four departures from the calibrated error model that were tested, the proposed model kept an advantage of 4.83 to 5.67 points over the nominal model. In software-in-the-loop flights with PX4 and Gazebo, the fallback raised the utility by 10.35 percentage points, whereas the advantage over the nominal model was not confirmed, as the kinematic tier had predicted for the same missions. An earlier study, in which only the stopping rule was made execution-aware, found no significant gain in utility (Section VI-A).

II. RELATED WORK

A. DYNAMIC TASK ALLOCATION AND ROUTING

Multi-robot task allocation [1], [21] and dynamic vehicle routing [2], [3] reassign tasks when information changes, and recent reviews cover their use for UAVs [4], [22]. Learning-based schedulers combine policy-gradient reinforcement learning with stable matching for dynamic multi-UAV task scheduling [23]. The planning problem considered here is a prize-collecting routing problem with time windows [24], [25]. Large neighbourhood search has been adapted to dynamic routing with time windows [26]. For multi-UAV systems, auction methods such as CBBA reach conflict-free assignments quickly [5] and have been extended

to dynamic task arrivals with time windows [27]. Recent work handles stochastic task completion with hierarchical policies [28], uncertain execution times with dynamic time slices [29], intermittent communication with uncertainty-gated distributed allocation [30], and events with assignment-preserving or local repair [31], [32]. Event-triggered and periodic allocation have been compared for multi-UAV systems [33], in line with the policies discussed for stochastic dynamic routing [3]. Calvo and Capitan validate a repair-based architecture with PX4 and Gazebo [34], and Ali et al. use ArduPilot software-in-the-loop simulation [35]. Table 1 compares these approaches.

B. UNCERTAINTY IN ROUTING AND ALLOCATION

Stochastic vehicle routing models travel and service times as random variables, including correlated and time-dependent travel times [9], [39], and multi-robot planning under uncertainty treats the durations of actions as random [40]. With hard time windows and stochastic service times, a priori routes with recourse [41] and robust formulations over budgeted uncertainty sets [7] protect the schedule against delays, at the price of a harder model. Sampling-based and learning-based policies plan against possible futures [3], and chance-constrained task allocation bounds the probability that a team misses its objectives [11]. Per-customer chance constraints on the service time windows were formulated for routing with stochastic travel and service times by Li et al. [42]. Ehmke et al. [8] propagate the mean and variance of the arrival time along a route, approximate the waiting for a time window by Clark's moments of a maximum [43], and use the result as a feasibility check inside routing heuristics. Soft time windows take the opposite route and charge a penalty for every late service inside the objective. Taş et al. [16] minimise transportation costs plus expected early and late service costs under stochastic travel times with tabu search, which requires the distribution of the lateness of every route during the search. For multi-UAV systems, Gao et al. [12] formulate a distributionally robust chance-constrained task assignment with time windows under uncertain fuel consumption and solve a mixed-integer reformulation, Wang et al. [38] estimate the effect of uncertain flight times on completion times and time-window violations without Monte Carlo simulation, and Zhou and Tokekar [36] select risk-averse multi-robot assignments through the conditional value at risk. For drone delivery, the energy and the range of a drone depend on its speed [44], Bruni et al. [37] protect routes against uncertain flight times by robust optimisation, and Cheng et al. [10] build ambiguity sets of flight times from wind observations and schedule drones against an index that accounts for both the probability and the magnitude of lateness. These formulations rely on mixed-integer, sampling-based or greedy solvers over a planning horizon, and none of them is evaluated as a replanner that has to publish a plan within a fraction of a second after every event.

The proposed model differs in four respects. Its error model is fitted to the finish times observed in closed loop rather than assumed per arc. It is applied in a replanner that runs between events within a fraction of a second, with a publication delay. It adds a fallback for the tasks that a hard chance constraint gives up. And it prices a late arrival only in this fallback, through a break-even probability that again yields a deterministic deadline, so the search keeps its objective and its fast feasibility checks, whereas soft time windows change both. The closest use of a probability as an acceptance test in multi-UAV allocation is the uncertainty-gated allocation of Qian and Fu [30], which removes an agent–task pair from a one-task-per-UAV matching when a lower confidence bound fails. There the test acts on single assignments under intermittent communication, whereas here it acts on the deadlines of every task along a route, with an error that grows with the lead time. Moment propagation, a proportional buffer, a constant buffer and the consensus-based bundle algorithm (CBBA) [5] are re-implemented as baselines.

C. METAREASONING AND ANYTIME ALGORITHMS

Anytime algorithms and metareasoning [13], [15] decide how long a planner should compute, from performance profiles [17], online performance prediction [14] or learned stopping policies [45]. Situated temporal planning and deadline-aware search account for the passage of time during planning [46], [47]. Two of these rules are re-implemented as baselines, so that the value of controlling computation can be compared with the value of the planning model.

III. PROBLEM FORMULATION

A central planner is called whenever a replanning trigger fires. Each UAV keeps the task that it is currently flying to or serving, which determines when and where it becomes available again and how much energy it has left. Each open task j has a location, a release time r_j, a deadline d_j by which its service must be completed, a service duration and a reward w_j. A plan assigns an ordered sequence of open tasks to each UAV, and tasks may remain unassigned, so the planning problem is a prize-collecting routing problem with time windows [24], [25]. Arrival, start and finish times along a route are computed as usual for time windows [48]. A route is feasible if every service ends before its deadline and the UAV returns to the depot with an energy reserve of 10% of its capacity. A plan π is evaluated by

$$J(\pi) = \alpha R(\pi) - \gamma E(\pi) - \delta C(\pi), \tag{1}$$

where R is the reward of the tasks served on time, E the flown distance in kilometres and C the number of tasks that move to another UAV than in the previous plan, with $\alpha = 1$, $\gamma = 0.05$ per kilometre and $\delta = 0.3$. W denotes the total reward of the open tasks.

Computation is not free. A plan computed for τ milliseconds is released $\kappa\tau$ milliseconds later in mission time, where the latency factor κ represents planning computers slower than the evaluation computer. An idle UAV waits for the new plan, so its route starts later, and tasks whose deadline can no

TABLE 1. Features of related approaches to task allocation, routing and replanning under uncertainty compared with the proposed method. TW: time windows or deadlines; Unc.: execution or duration uncertainty in the planning model; Late cost: a cost of late service that enters the planning decision; Online: replanning during the mission; Trigger: replanning trigger policies compared; Stop: when to stop the replanner (computation as a decision); SITL: validation with an autopilot in software-in-the-loop or physics simulation. ✓: yes; ○: partly; –: no or not reported.

Work	Problem	TW	Unc.	Late cost	Online	Trigger	Stop	SITL	Solver or architecture
Taş et al. [16]	VRP, soft time windows	✓	✓	✓	–	–	–	–	tabu search
Ehmke et al. [8]	VRPTW, service levels	✓	✓	–	–	–	–	–	heuristic, moments
Zhou and Tokekar [36]	multi-robot assignment	–	✓	–	–	–	–	–	greedy, CVaR
Cheng et al. [10]	drone scheduling, wind	✓	✓	✓	○	–	–	–	DRO, branch and cut
Bruni et al. [37]	drone location routing	–	✓	–	–	–	–	–	robust optimisation
Gao et al. [12]	UAV assignment	✓	✓	–	–	–	–	–	MILP, DRCC
Wang et al. [38]	UAV task planning	✓	✓	–	–	–	–	–	completion-time estimate
Choudhury et al. [28]	MRTA, time windows	✓	✓	–	✓	–	–	–	hierarchical
Huang et al. [29]	multi-UAV allocation	–	✓	–	✓	○	–	–	–
Calvo and Capitan [34]	long-endurance UAV MRTA	○	–	–	✓	–	–	✓	centralised
Hwang et al. [31]	multi-UAV replanning	✓	–	–	✓	–	–	–	–
Shen et al. [32]	UAV swarm allocation	○	–	–	✓	–	–	–	edge
Qian and Fu [30]	multi-UAV allocation	✓	✓	–	✓	–	–	–	distributed
Cong et al. [33]	multi-UAV allocation	✓	○	–	✓	✓	–	○	–
Ali et al. [35]	multi-UAV allocation	–	–	–	✓	–	–	✓	decentralised
Svegliato et al. [14]	anytime metareasoning	–	–	–	–	–	✓	–	–
Proposed	multi-UAV replanning	✓	✓	✓	✓	✓	✓	✓	centralised

longer be met are lost. The plan then reaches the vehicles after a dissemination latency L that the planner does not control.

The planner uses a nominal model with straight legs at the cruise speed of each UAV and nominal service durations. The executed finish time of task j is $F_j = f_j + e_j$, where f_j is the nominal finish time at publication and e_j an execution error caused by acceleration, wind, speed tracking and service-time variability. A task yields its reward only if $F_j \le d_j$. As in dispatching practice, a UAV skips a task at departure if the nominal estimate from its actual state already misses the deadline, and the task returns to the pool. The utility of a mission is the reward of all tasks served on time divided by the total reward of the mission. Three replanning triggers are considered. The event trigger replans on task arrivals, UAV failures and deadline changes, the periodic trigger in addition every T seconds, and the deviation trigger in addition when a finish time deviates from the plan by more than θ seconds. During planning, every UAV finishes its committed task and then waits, and events are queued.

TABLE 2. Notation.

Symbol	Meaning
t_0	time of the planning snapshot
r_j, d_j, w_j	release time, deadline and reward of task j
f_j, F_j	planned (nominal) and actual finish time of task j
$h_j = f_j - t_0$	lead time of task j
$e_j = F_j - f_j$	finish-time error, $e_j \sim \mathcal{N}(m(h_j), s(h_j)^2)$
$m(h)$, $s(h)$	mean and spread of the error, $b_0 + b_1 h$ and $s_0 + s_1 h$
z	safety factor; protected tasks meet d_j with probability $\ge \Phi(z)$
$\tilde{d}_j(z)$	effective deadline, Eq. (4)
$\tilde{H}$, H	tasks no UAV can finish by $\tilde{d}_j$, resp. by d_j, on a direct flight
c_j, λ	cost of a late arrival at task j; λ: its size in task rewards
θ_j, z_j	break-even probability $c_j/(w_j + c_j)$ and $\Phi^{-1}(\theta_j)$
d'_j	planning deadline used by repair and search
τ, B	planning time of a call and its hard budget
κ, L	latency factor and communication latency
u_λ, u^w_λ	scores $U - \lambda n_{\text{late}}/n$ and $U - \lambda R_{\text{late}}/R$

IV. PROPOSED METHOD

LTCC (lead-time-scaled chance constraints) takes four decisions in every replanning cycle (Fig. 1, Algorithm 1). The chance-constrained planning model decides which plans are acceptable under the calibrated execution error. The cost-aware fallback decides which of the tasks that this model gives up are still worth attempting, given the cost of a late arrival. The stopping rule decides when to stop computing and publish, and the trigger decides when to replan. The first two decisions are the contribution of this paper. Both reduce to one effective deadline per task, so a feasibility-first repair and an anytime ALNS, or any other heuristic that checks deadlines, produce the plan unchanged. Table 2 lists the notation. Two earlier variants serve as baselines. AnytimeSwarm uses the same stopping rule with a deterministic delay cost on the nominal model, and AnytimeSwarm-R uses an execution-aware delay cost on the nominal model.

A. REPAIR AND ANYTIME SEARCH

The previous plan is restricted to the open tasks and walked once, keeping a task only if the route up to it is still feasible, and tasks of failed UAVs and new tasks are inserted by parallel regret-2 insertion [6]. This feasibility-first repair yields a publishable plan within milliseconds. ALNS [6] then improves it with four destroy operators (random, worst, related and route-segment removal), four greedy or regret-2 repair operators with and without noise, adaptive operator selection and simulated-annealing acceptance, the standard components of ALNS [6], [49]. All intermediate solutions are feasible, so the best plan found so far can be published at any time. The temperature decreases per iteration and all variants share the random seed, so every stopping rule follows the same search trajectory and differs only in the moment at which it stops.

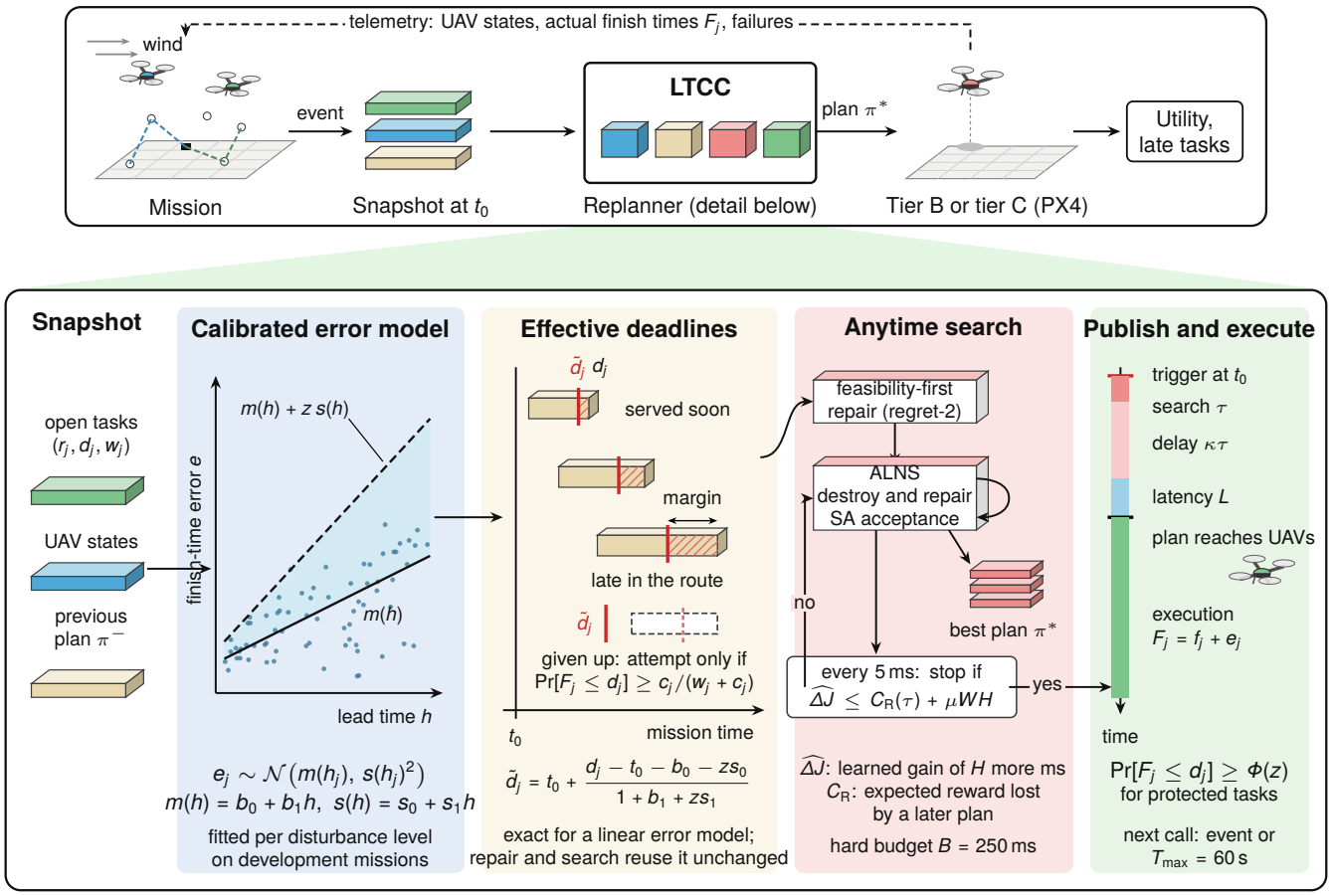


FIGURE 1. **Overview of LTCC. Top: the replanning loop between the mission, the planning snapshot, the replanner and the execution tier. Bottom: one replanning call. The calibrated error model gives a margin that grows with the lead time, the chance constraint turns it into an effective deadline per task, the cost-aware fallback keeps a given-up task only if its probability of meeting the deadline reaches the break-even value $c_j/(w_j + c_j)$ set by the reward w_j and the cost c_j of a late arrival, and the anytime search stops when the predicted gain no longer pays for the delay. The plan is released after the emulated planning delay $\kappa\tau$ and the latency L.**

B. CALIBRATED FINISH-TIME ERROR

The error of task j is modelled as $e_j \sim \mathcal{N}(m(h_j), s(h_j)^2)$, where $h_j = f_j - t_0$ is its lead time, the nominal finish time minus the time t_0 of the planning snapshot, and

$$m(h) = b_0 + b_1 h, \qquad s(h) = s_0 + s_1 h, \tag{2}$$

because errors accumulate along a route. The parameters are fitted per disturbance level on development missions, the mean by least squares and the spread by least squares on the absolute residuals scaled by $\sqrt{\pi/2}$, which is less sensitive to the heavy tail of the errors. Under the mid and high disturbances of the main studies, both the mean and the spread grow with the lead time (Supplementary Table S2). The nominal model is therefore optimistic, and increasingly so along a route.

The affine spread is a modelling choice that the data can reject. A route-level model with independent increments per unit of time, the usual assumption when variances are added along a route, implies $s(h) = \sqrt{q_0 + q_1 h}$. On 4 982 and 4 231 pairs of lead time and error from 90 development missions per level, the affine spread fits better by 859 and 910 points of the Akaike information criterion under mid and high disturbances, and also better than the hybrid $\sqrt{q_0 + q_1 h + q_2 h^2}$ (by 72 and 117). A five-fold cross-validation with missions as groups gives the same ranking. At the longest lead times, about 514 s, the square-root model predicts a spread of 262 s under high disturbances where 372 s are observed, while the affine model predicts 380 s (Fig. 2a).

The reason is that the disturbances persist over a leg. Wind and an airspeed error act during the whole flight to a task, so the error of a leg is proportional to its duration (Fig. 2b), with a coefficient of variation of 0.39 and 1.15 under mid and high disturbances. Errors of successive legs are close to independent. The correlation between the error of a task and the error added by the next task of the same plan was -0.20 and -0.05, and missions explained only 5% and 7% of the residual variance. A lead time of a few legs therefore carries a spread that grows almost linearly. Moment propagation with a variance proportional to the squared duration of each leg is consistent with this structure and serves as the literature baseline in Section V-C.

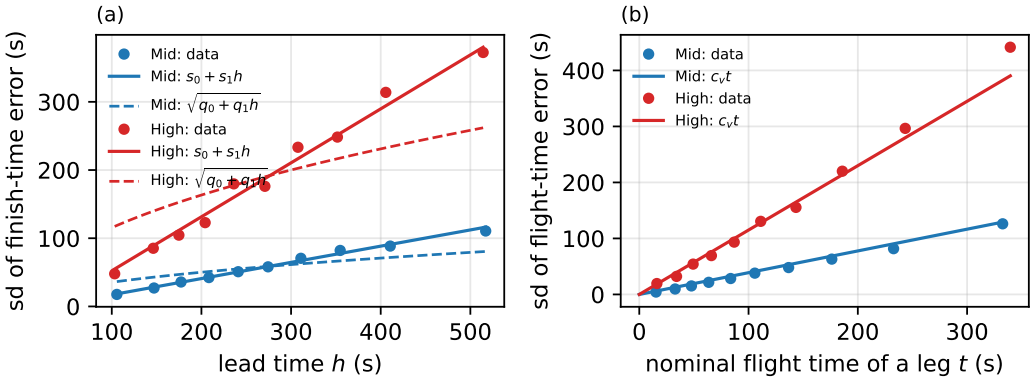


FIGURE 2. Structure of the finish-time error on development missions. (a) Standard deviation of the error against the lead time, in deciles, with the affine fit of Eq. (2) and the fit with independent increments. (b) Standard deviation of the flight-time error of single legs against their nominal flight time, in deciles, with a spread proportional to the duration.

Algorithm 1 One replanning call of LTCC

Require: snapshot at time t_0: committed tasks, open tasks with (r_j, d_j, w_j), UAV states; previous plan π^-; error model (b_0, b_1, s_0, s_1), safety factor z, late costs c_j, expected latency $\bar{L}$, latency factor κ

$z_j \leftarrow \Phi^{-1}(c_j/(w_j + c_j))$; $z_{p,j} \leftarrow \max\{z, z_j\}$ for every open task ▷ $z_j = -\infty$ for $c_j = 0$

$\tilde{d}_j(\cdot) \leftarrow \min\{d_j,\ t_0 + (d_j - t_0 - b_0 - \cdot\, s_0)/(1 + b_1 + \cdot\, s_1)\}$ for every open task ▷ Eq. (4)

$d'_j \leftarrow \min\{d_j, \tilde{d}_j(z_j)\}$ if no UAV can finish j by $\tilde{d}_j(z_{p,j})$ but one can by d_j, else $d'_j \leftarrow \tilde{d}_j(z_{p,j})$ ▷ Eq. (7)

$\pi \leftarrow$ FeasibilityFirstRepair(π^-, d'); $\pi^* \leftarrow \pi$; $\tau \leftarrow 0$

repeat

  destroy and repair π with adaptively chosen operators; accept by simulated annealing

  if $J(\pi) > J(\pi^*)$ **then** $\pi^* \leftarrow \pi$

  every 5 ms: $\widehat{\Delta J} \leftarrow$ learned prediction of the gain in the next H ms

until $\widehat{\Delta J} \le C_{\mathrm{R}}(\tau) + \mu W H$ **or** $\tau \ge B$ ▷ Eq. (8)

publish π^*; it reaches the UAVs at $t_0 + \kappa\tau + L$

next call at the next event or $T_{\max}$ after t_0, whichever comes first

C. CHANCE-CONSTRAINED PLANNING MODEL

The proposed model plans for the calibrated execution. A planned finish time f_j is accepted only if

$$f_j + m(f_j - t_0) + z\, s(f_j - t_0) \;\le\; d_j, \tag{3}$$

where $z \ge 0$ is a safety factor. The margin $m + zs$ depends on the finish time itself, so it is small for tasks served soon and large for tasks served late in a route. Because m and s are linear, the constraint reduces to an effective deadline,

$$f_j \le \tilde{d}_j, \qquad \tilde{d}_j = t_0 + \frac{d_j - t_0 - b_0 - z s_0}{1 + b_1 + z s_1}. \tag{4}$$

The repair and the search use $\tilde{d}_j$ instead of d_j, while the executor and the reward still use the true deadline.

Proposition 1: Let $1 + b_1 + z s_1 > 0$ and $s_0, s_1 \ge 0$, so that $s(h) \ge 0$ for every lead time $h \ge 0$. (i) A finish time satisfies Eq. (3) if and only if $f_j \le \tilde{d}_j$. (ii) Under the error model of Eq. (2), a task planned to finish at $f_j \le \tilde{d}_j$ meets its deadline with probability at least $\Phi(z)$ if its route starts as planned. (iii) With $b_0 = b_1 = s_0 = s_1 = 0$, $\tilde{d}_j = d_j$ and the planning model is the nominal one.

Proof: Substituting Eq. (2) into Eq. (3) gives $f_j(1 + b_1 + z s_1) \le d_j - b_0 - z s_0 + (b_1 + z s_1) t_0$, and dividing by the positive factor gives (i). For (ii), the actual finish is $F_j = f_j + e_j$ with $e_j \sim \mathcal{N}(m(h_j), s(h_j)^2)$ and $h_j = f_j - t_0$. By (i), $d_j - f_j - m(h_j) \ge z\, s(h_j)$. If $s(h_j) > 0$, then $\Pr[F_j \le d_j] = \Phi\big((d_j - f_j - m(h_j))/s(h_j)\big) \ge \Phi(z)$. If $s(h_j) = 0$, which happens only for $s_0 = 0$ and $h_j = 0$, the error equals $m(h_j)$ with certainty, $F_j \le d_j$ holds, and the probability is one. Part (iii) is immediate. ■

The guarantee holds under the stated error model. It is only as good as the calibration of that model, which Section VI-D and Supplementary Section S9 check on held-out missions, and study 7 tests errors that the model does not describe. The planner uses $\min\{d_j, \tilde{d}_j\}$, so that no task is planned beyond its nominal deadline. Where $\tilde{d}_j > d_j$, which happens when $z < 0$ or the mean error is negative, the cap adds the nominal deadline to Eq. (3). The constraint then becomes tighter, so part (ii) still holds, whereas the equivalence of part (i) refers to the uncapped deadline.

The effective deadline turns a chance constraint on every task into a deterministic constraint of the same form as the original one. The insertion heuristics and the feasibility checks of the search are therefore reused unchanged, and computing $\tilde{d}_j$ costs one division per task. With $z = 0$, the model corrects the mean error only and plans for the median finish time. Larger values of z protect more tasks against lateness but leave fewer tasks feasible, so z trades the risk of a late service against the reward of attempting a task. It is chosen per disturbance level on development missions.

D. NOMINAL FALLBACK FOR GIVEN-UP TASKS

A hard chance constraint has a side effect when the time windows are short compared with the spread of the error. If no vehicle can finish task j before $\tilde{d}_j$, even on a direct flight, the task is infeasible for the planner and is never attempted. A vehicle may still serve it in time with a probability below

$\Phi(z)$ but well above zero, and it may have nothing better to do. The fallback keeps such tasks. Let $\tilde{H}$ be the set of tasks that no vehicle can finish by $\tilde{d}_j$ on a direct flight, and H the set of tasks that no vehicle can finish by d_j. The planning deadline is

$$d'_j = \begin{cases} d_j, & j \in \tilde{H} \setminus H, \\ \tilde{d}_j, & \text{otherwise.} \end{cases} \tag{5}$$

The constraints of all other tasks are unchanged, so every other task that is served keeps the guarantee of Proposition 1, while a fallback task carries none. Finding $\tilde{H}$ and H requires one direct-flight check per task and vehicle, which is negligible compared with the search. The fallback was designed after the held-out results of study 2 and is evaluated on new seeds in studies 3 and 4.

E. COST-AWARE FALLBACK

The nominal fallback is right only if a late service costs nothing. In many applications a late arrival has a cost of its own. The flight time and energy are spent, a customer may have left, or a commitment is broken. Let a late arrival at task j cost $c_j \geq 0$. An attempt that meets the deadline with probability p then has the expected value $p\,w_j - (1-p)\,c_j$, which is non-negative if and only if

$$p \;\geq\; \theta_j = \frac{c_j}{w_j + c_j}. \tag{6}$$

Two cost models are natural, and both measure the cost by a factor λ in units of a task reward. If a late arrival forfeits a share of the value of the task, $c_j = \lambda w_j$ and $\theta_j = \lambda/(1+\lambda)$ for every task. If it costs a fixed amount, such as a wasted flight or a contractual penalty, $c_j = \lambda \bar{w}$ with the mean task reward $\bar{w}$, and $\theta_j = \lambda\bar{w}/(w_j + \lambda\bar{w})$ is lower for valuable tasks. By Proposition 1, a finish time $f_j \leq \tilde{d}_j(z)$ gives a probability of at least $\Phi(z)$, where $\tilde{d}_j(z)$ denotes the effective deadline of Eq. (4) at safety factor z. The break-even probability of task j therefore corresponds to the safety factor $z_j = \Phi^{-1}(\theta_j)$. The cost-aware fallback protects task j at $z_{p,j} = \max(z, z_j)$, and plans a task that no vehicle can finish by $\tilde{d}_j(z_{p,j})$ on a direct flight against $\min\{d_j, \tilde{d}_j(z_j)\}$. A task that cannot meet this deadline either is given up. Thus

$$d'_j = \begin{cases} \min\{d_j, \tilde{d}_j(z_j)\}, & j \in \tilde{H}_p \setminus H, \\ \tilde{d}_j(z_{p,j}), & \text{otherwise,} \end{cases} \tag{7}$$

with $\tilde{H}_p$ the set of tasks that no vehicle can finish by $\tilde{d}_j(z_{p,j})$ on a direct flight. Each task keeps a closed-form deadline, so the cost model changes only the safety factor per task.

Proposition 2: Let $s_0, s_1 \geq 0$, let $1 + b_1 + z s_1 > 0$ and $1 + b_1 + z_j s_1 > 0$ for every finite z_j, and read $\tilde{d}_j(-\infty)$ as d_j. (i) If $c_j = 0$ for every task, Eq. (7) is the nominal fallback of Eq. (5). (ii) A task with $\theta_j \geq \Phi(z)$ never falls back, and it is planned against $\tilde{d}_j(z_j)$. (iii) Local break-even: under the error model of Eq. (2), every task planned to finish by d'_j has an expected contribution $w_j \Pr[F_j \leq d_j] - c_j \Pr[F_j > d_j] \geq 0$ if its route starts as planned. (iv) If every c_j is non-decreasing in λ, every planning deadline d'_j is non-increasing in λ.

Proof: (i) $z_j = -\infty$ gives $z_{p,j} = z$ and $\min\{d_j, \tilde{d}_j(-\infty)\} = d_j$. (ii) $z_j \geq z$ holds if and only if $\theta_j \geq \Phi(z)$. Then $z_{p,j} = z_j$, and a task that cannot meet $\tilde{d}_j(z_{p,j})$ cannot meet $\min\{d_j, \tilde{d}_j(z_j)\}$ either, so it is given up. (iii) A protected task meets its deadline with probability at least $\Phi(z_{p,j}) \geq \Phi(z_j)$, and a fallback task with probability at least $\Phi(z_j)$, both by Proposition 1, whose part (ii) also holds with the cap at d_j. Hence $p \geq \theta_j$, and the expected contribution $(w_j + c_j)p - c_j$ is not negative. (iv) θ_j and z_j do not decrease with λ, and with $s_0, s_1 \geq 0$, $\tilde{d}_j(z)$ does not increase with z. A fallback task stays one when λ grows, because $z_{p,j}$ does not decrease. A protected task can become a fallback task only while $z_j < z$, and then $z_{p,j} = z$ does not change. In each case the deadline does not increase. ■

Part (iii) is a local criterion. It prices each attempt on its own and ignores the delay that a fallback task causes to the tasks after it on the same route, as well as the reassignments it may trigger, so it does not guarantee that attempting the task raises the utility of the whole plan. With this caveat, the rule attempts a given-up task only if its expected contribution is not negative. It adds no parameter to tune, because z keeps its value and the late cost is a property of the application. Where $1 + b_1 + z_j s_1 \leq 0.05$, the effective deadline is not defined and the nominal deadline is used, without the guarantee of part (iii). With the error models of this paper this happens only for $\theta_j < 0.03$, which none of the studied late costs produces. The rule needs only a probability of meeting the deadline, so it applies to any planning model that provides one, such as moment propagation, where the given-up tasks are accepted if $\mathrm{E}[F_j] + z_j \mathrm{sd}[F_j] \leq d_j$. A margin without a probability of meeting the deadline, such as a proportional buffer, has no counterpart of z_j, so its fallback deadline has to be tuned for every late cost. The cost-aware fallback was designed after the held-out results of study 4 and is evaluated on new seeds in studies 5 to 7. Studies 5 and 6 ran it with the proportional cost model, $\theta_j = \lambda/(1+\lambda)$, but scored the runs with a fixed cost per late arrival, u_λ. Because the rewards differ between tasks, the rule was thus derived for another cost than the one it was scored with. Study 7 runs both cost models, each scored with its own metric.

F. STOPPING RULE AND TRIGGER

Every 5 ms, the stopping rule compares the predicted gain $\widehat{\Delta J}$ of searching $H = 10$ ms longer with the cost of doing so, and stops at time τ if

$$\widehat{\Delta J} \;\leq\; C_{\mathrm{R}}(\tau) + \mu W H, \tag{8}$$

with a price of computation $\mu = 10^{-5}$ per millisecond. The gain is predicted from eleven features of the search history by a Poisson generalised linear model [50] fitted on development trajectories. The delay cost $C_{\mathrm{R}}(\tau) = \Lambda_{\mathrm{R}}(\tau + H) - \Lambda_{\mathrm{R}}(\tau)$ is the increase of the expected value of the tasks that a later start would lose,

$$\Lambda_{\mathrm{R}}(\tau) = \sum_j v_j \, \Phi\!\left(\frac{\sigma_j(\tau) + \bar{L} + m(h_j) - \mathrm{tol}_j}{s(h_j)} \right), \tag{9}$$

where $\sigma_j(\tau)$ is the start shift of the route of task j caused by waiting for the plan, tol_j the margin of the task to its deadline plus the waiting times before it on its route [48], $\bar{L}$ the expected latency and v_j the value of the task in Eq. (1). On the effective deadlines, the mean error is already part of the tolerance, so LTCC uses $m \equiv 0$. Λ_{R} does not decrease with τ, and it reduces to the exact loss of a deterministic plan when the spread vanishes (Supplementary Section S1). A hard budget B bounds every call. The replanner replans on every event and at least every $T_{\max} = 60\,\mathrm{s}$ of mission time, a rolling-horizon policy in the terminology of dynamic routing [3].

V. EVALUATION

A. SCENARIOS AND KINEMATIC TIER

Missions are created by a seeded generator (Supplementary Table S1). The map is a square with a side of 10 km and the depot at its centre, half of the tasks are clustered around three hotspots, and half of them are known at the start while the others arrive in batches. Time windows have a slack of 180 to 600 s and services last 30 to 90 s. Task rewards are 1, 2, 3 or 5 with probabilities 0.4, 0.3, 0.2 and 0.1, so the mean reward is $\bar{w} = 2.1$. Three fleet scales with 5, 10 and 20 UAVs and 25, 50 and 140 tasks are combined with three event profiles, namely six arrival batches, three batches with two UAV failures, and three batches with four tightened deadlines.

Tier B executes each published plan in continuous mission time. Legs follow a trapezoidal speed profile with an acceleration limit, the airspeed of each leg deviates from the nominal speed by a multiplicative error, a constant wind of random direction changes the ground speed, and service durations are multiplied by a random factor (Table 3). The planner never sees these disturbances and receives the state of each UAV as telemetry. The dissemination latency is lognormal, and lost messages are resent after a timeout. All random draws use common random numbers [51] keyed to the mission seed, the UAV and the task, so two planners that assign the same task to the same UAV experience the same disturbance.

B. SOFTWARE-IN-THE-LOOP TIER

Tier C flies missions with three X500 quadrotors, each controlled by its own PX4 v1.17 autopilot in software-in-the-loop mode [18], in Gazebo Harmonic [19] on ROS 2 Jazzy [20] (Fig. 3). The mission manager runs the scenario timeline, builds the planning snapshot, calls the planner and releases the plan. The executor streams offboard setpoints to every autopilot at 10 Hz through a uXRCE-DDS bridge, flies each leg with a carrot that moves along the straight leg and holds the UAV at the task for the service duration. A tier C mission is a tier B scenario with three UAVs and fifteen tasks, rescaled by $S_X = 0.1$ in length and $S_T = 0.25$ in time. Speeds are multiplied by S_X/S_T, energy by S_X and the distance price by $1/S_X$, so every dimensionless ratio of the decision problem is preserved, and $\kappa_C = S_T \kappa_B$. The commanded ground speed of each leg follows the speed error and the scaled wind of tier B with the same random numbers, while acceleration, attitude dynamics, position tracking and communication are those of the simulated vehicles. Gazebo runs in lockstep with PX4 at twice real time, and mission time is simulation time. Every task is drawn in the Gazebo scene as a pillar whose colour shows its outcome, and the PX4 logs record the flown trajectories (Fig. 4).

C. COMPARED METHODS

All methods share the feasibility-first repair, the ALNS operators and parameters, the initial plan and the random seed, and every call has a hard budget of 250 ms. They differ in the stopping rule, the planning model and the trigger. Regret-2 repair without search is the constructive floor. ALNS is run with fixed budgets of 250 ms and 40 ms [6], the latter equal to the mean computation of AnytimeSwarm on development data, with a budget of 1 ms per open task, and with a stagnation rule that stops after 20% of the budget without improvement. Two stopping rules from the metareasoning literature are re-implemented with the same delay cost and price of computation as the proposed rule, so that only the prediction of the gain differs. The performance profile rule [13], [17] predicts the gain from a static profile over elapsed time and number of open tasks, and the online prediction rule [14] fits $q(t) = a - b\,e^{-ct}$ to the best objective found so far at every check. The constant-buffer model subtracts 60 s from every deadline, chosen on development missions among 30, 60 and 120 s, as is common for robust schedules with time buffers [7]. The proposed model uses Eq. (4) with $z = 0.5$ under mid and $z = 0$ under high disturbances, selected on development missions. All these methods use the periodic trigger. The conventional planner combines ALNS with 40 ms, the nominal model and the event trigger.

Study 4 compares planning models on the same search, with a fixed budget of 40 ms, the periodic trigger and the nominal fallback for every chance-constrained model. The proportional buffer accepts $f_j + c\,h_j \le d_j$, which keeps the shape of the proposed margin but replaces the calibration by one tuned slope. Moment propagation follows Ehmke et al. [8]. The flight time of a leg of nominal duration t is normal with mean $(1+a)t+c_0$ and standard deviation $c_v t$, the service time is normal with mean $(1 + a_s)s$ and standard deviation $c_s s$, legs are independent, the start of service $\max(A_j, r_j)$ is approximated by Clark's moments [43], and a finish is accepted if $\mathrm{E}[F_j]+z\,\mathrm{sd}[F_j] \le d_j$. The leg parameters are fitted on the legs of the development flights. Because the variance of one long leg can exceed that of the two legs it replaces, removing a task can make a route infeasible under this model, so every route is filtered once more before publication. Online calibration starts from one error model fitted on both disturbance levels, with the weight of n_0 pairs, and refits Eq. (2) at every call with the pairs of the tasks completed so far in the mission, so it needs neither the disturbance level nor a level-specific z. Each model has one parameter per level, selected on development missions by the same rule as z in study 2. For the proposed model with fallback, z was selected anew and is 1 under mid and 0.5 under high disturbances. The other

TABLE 3. Disturbance and latency levels of tier B. The levels span mild to severe conditions and are not claimed to represent a particular vehicle.

Execution	accel. (m/s^2)	speed c.v.	wind (m/s)	service c.v.	Latency	median (s)	$\sigma_{\log}$	loss / timeout
low	3.0	0.03	2	0.05	low	0.2	0.5	–
mid	2.0	0.07	5	0.15	high	2.0	0.8	5% / 5 s
high	1.5	0.12	8	0.25				

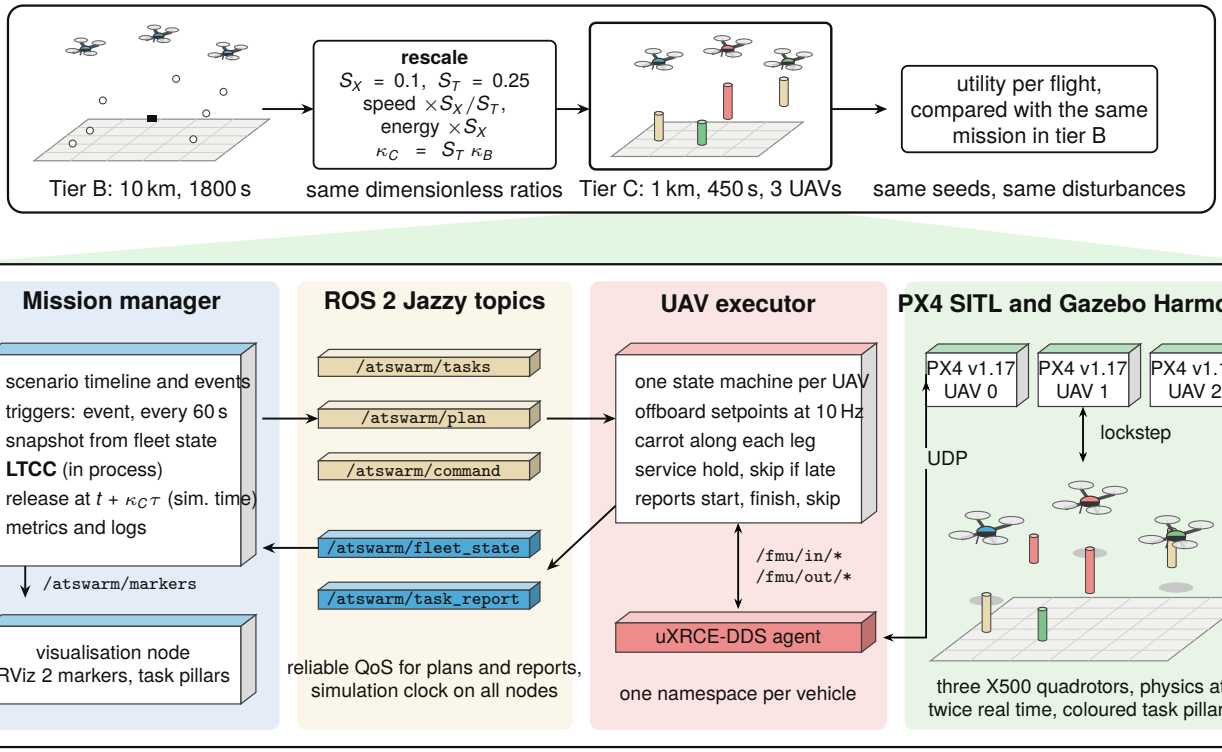


FIGURE 3. Software-in-the-loop tier. Top: a tier B mission is rescaled in length and time so that the decision problem is unchanged. Bottom: the planner runs inside the ROS 2 mission manager, the executor streams offboard setpoints through the uXRCE-DDS bridge to three PX4 autopilots, and Gazebo simulates three X500 quadrotors in lockstep.

values are $c = 0.45$ and 0.6, $z = 1$ and 0.5 for moment propagation, and $n_0 = 50$ with $z = 0.5$ for online calibration. On development missions, the three static models differed by less than 0.2 pp.

Study 5 charges every late arrival with λ times the mean task reward of the mission and scores a run by $u_\lambda = U - \lambda\, n_{\text{late}}/n$, where U is the utility, n_{late} the number of tasks served late and n the number of tasks. $\lambda = 0$ gives the utility of studies 1–4. The cost-aware fallback of Section IV-E is run with the λ it is scored with and the z of study 4, without any further tuning, and with the proportional cost model, $\theta_j = \lambda/(1+\lambda)$. Because U is a share of the total reward, the score u_λ instead charges every late arrival the fixed cost $\lambda\bar{w}$, a mismatch that study 7 removes. Its competitors are tuned for every $\lambda \in \{0.5, 1, 2\}$ and disturbance level on development missions. These are the proportional buffer with $c \in \{0.1, 0.2, 0.3, 0.45, 0.6, 0.75, 0.9, 1.2\}$, with or without the nominal fallback, and the proposed model without fallback with $z \in \{-0.5, 0, 0.5, 1, 1.5\}$. A grid that first ended at $c = 0.3$ and $z = 0$ was extended after both values were selected on its edge, and the selection did not change. For $\lambda = 1$, the selected buffer was $c = 0.3$, no fallback under mid and $c = 0.45$, no fallback under high disturbances, and the selected model without fallback had $z = 1$ and $z = 0.5$.

Study 6 adds a published task-allocation algorithm and applies the cost-aware fallback to moment propagation. The consensus-based bundle algorithm (CBBA) [5] is re-implemented with bids equal to the largest increase of the route value over all feasible insertion positions, a fully connected network, and the bid warping of Johnson et al. [52]. Time windows break the diminishing marginal gains on which the convergence of CBBA rests, and without bid warping it did not converge on some development snapshots. CBBA plans from scratch at every call, stops at convergence or at the hard budget, and pays for its planning time like every other method. It runs with the periodic and with the event trigger, and also on the deadlines of the cost-aware fallback with $\lambda = 1$. Moment propagation with the cost-aware fallback accepts a given-up task if its nominal finish meets the deadline and $\mathrm{E}[F_j] + z_j \mathrm{sd}[F_j] \le d_j$. No parameter

was tuned in study 6.

Study 7 tests two assumptions of the method. Study 7a removes the mismatch between the cost model of the rule and the score of studies 5 and 6. The cost-aware fallback runs with the proportional cost model and with the fixed cost model, $\theta_j = \lambda\bar{w}/(w_j + \lambda\bar{w})$ with the mean reward $\bar{w} = 2.1$ of the generator, and every run is scored with both u_λ and $u^w_\lambda = U - \lambda R_{\text{late}}/R$, where R_{late} is the reward of the tasks served late and R the total reward, so that each cost model has its own score. The fixed-cost rule uses the mean reward $\bar{w}$ of the generator, because the total reward of a mission is not known while tasks are still arriving, whereas u_λ charges a late arrival the realised mean reward R/n of the mission. Over the held-out missions, R/n was 2.10 on average, with a coefficient of variation of 8% and a range of 1.64 to 2.76. Study 7a is therefore also reported with $u'_\lambda = U - \lambda\bar{w}\, n_{\text{late}}/R$, which charges exactly the cost that the rule assumes. The proportional buffer and the model without fallback are tuned again on development missions for every λ, disturbance level and score, from the candidates of study 5. Study 7b keeps the error model, z and every tuned parameter of the calibrated high level, but draws the execution errors from structures that the model does not describe. Speed and service errors follow a Student-t law with three degrees of freedom at the same standard deviation; rare long delays add, with probability 0.1, an exponential delay with a mean of 90 s to a leg, as a detour or a holding pattern would; a persistent speed bias per UAV and mission with a standard deviation of 8% correlates the errors along a route; and a Pareto dissemination delay with the median of the lognormal one and a tail index of 1.5 replaces the lognormal delay. The calibrated Gaussian level serves as control. Study 7b compares the nominal model, the constant and the proportional buffer, the proposed model with the nominal and with the cost-aware fallback ($\lambda = 1$, both cost models) and moment propagation with the cost-aware fallback.

D. PROTOCOL

The evaluation consists of seven studies, each with its own held-out seeds, written protocol and frozen code, and each study after the first was designed after the held-out results of the earlier ones. Table 4 lists their primary hypotheses. Development used seeds 0–9 in tier B and 0–2 in tier C, on which the predictor, the compute price, z, the buffers and the error models were fixed. Before each study, the written protocol with its hypotheses, the configurations, the models and the code were frozen with SHA-256 hashes and committed to the version history, and the held-out runs checked the hashes before starting. The protocols were not lodged with an external registry. They are published with the code, together with the commit of every freeze and every deviation.

Study 1 tested AnytimeSwarm-R against fixed and adaptive stopping rules on the nominal model (Supplementary Section S4). Studies 2–6 share one tier B matrix. It crosses two disturbance levels (mid, high), high latency, three latency factors ($\kappa_B = 10, 50, 200$) and two time-window factors (original and halved), which gives twelve configurations of 270 held-out missions each, with new seeds 5000–5029, 7000–7029, 8000–8029, 10000–10029 and 12000–12029 for studies 2, 3, 4, 5 and 6, and 13000–13029 and 14000–14029 for studies 7a and 7b. Study 2 compares fifteen methods (Supplementary Table S6), and its tier C part flies 30 missions with mid disturbances and $\kappa_B = 50$ (seeds 6000–6009). Study 4C flies 36 missions under high disturbances with halved windows, where tier B predicted the largest effect (seeds 9000–9011). Its size was computed beforehand from a tier B counterpart on development seeds, and one interim analysis after half of the missions uses O'Brien–Fleming bounds, so that the flights could stop early without inflating the error rate. Studies 5 and 6 test their primary hypotheses with the score u_1, study 7a with u_1 and u^w_1, and study 7b with U and u_1.

In all studies, the utility difference of two methods is averaged per mission over the configurations and tested with a paired bootstrap [53] of 10 000 resamples over missions. Primary families are corrected with the Holm procedure [54]. Non-inferiority uses a margin of -0.5 pp and is followed by a superiority test in a fixed sequence, and equivalence is tested with two one-sided tests. All other comparisons are descriptive. The nine missions of one seed have independent scenarios but share the execution disturbances, which are keyed to the seed. Every tier B primary test of studies 2–7 was therefore repeated with a paired bootstrap over the 30 seeds (Supplementary Table S10). The intervals were 0.69 to 1.18 times as wide as over missions, mostly narrower, because every seed contains one mission of each fleet scale and event profile. Only the outcome of H9 changed. There, the advantage of the proportional buffer over the proposed model became significant (Holm-adjusted $p = 0.032$). A tier C run in which a vehicle does not take off in time, or whose real-time factor leaves $[1.6, 2.2]$, is repeated once. Tier B runs four missions in parallel on an Intel Core i7-11800H laptop, with the methods of a configuration run in turn. A separately written validator found no infeasible plan among 4 779 development plans. The protocols and their deviations are available with the code.

VI. RESULTS

Table 4 summarises the primary hypotheses of studies 2–7, as fixed in their frozen protocols, and their outcomes. The subsections report each study in the order in which it was designed, so that every design decision can be traced to the results that motivated it.

A. STUDY 1 IN BRIEF: EXECUTION-AWARE STOPPING ON THE NOMINAL MODEL

Study 1 made only the stopping rule execution-aware and kept the nominal planning model. In 34 560 tier B runs, AnytimeSwarm-R needed 15% less computation per call than AnytimeSwarm, but its advantage over ALNS with 40 ms was +0.32 [+0.01, +0.63] percentage points (pp) with a Holm-adjusted p of 0.128. Periodic replanning raised the utility

TABLE 4. Primary hypotheses of studies 2–7 on held-out missions, as fixed in the frozen protocols, and their outcomes. Differences are the first method minus the second, in percentage points with 95% paired bootstrap intervals (for H27 and H28 the range over the four conditions); U is the utility, u_1 the score with a fixed late cost of one mean task reward and u_1^w the score with a late cost of one reward of the task. Primary families are Holm-corrected; NI: non-inferiority with a margin of -0.5 pp; sup.: superiority tested after NI in a fixed sequence. With a bootstrap over seeds instead of missions, the proportional buffer was significantly better than the proposed model in H9 (Supplementary Table S10); no other outcome changed. Study 1 is reported in Supplementary Section S4.

Study	H	Comparison	Score	Difference (pp)	Outcome
2	H1	proposed, no fallback, vs. six baselines	U	+1.35 to +2.87	supported, 6 of 6
	H2	proposed, no fallback, vs. conventional	U	+4.44 [+3.82, +5.05]	supported
	H3	stopping rule vs. fixed 40 ms (equivalence)	U	+0.22 [+0.07, +0.37]	equivalent
	H4	tier C: proposed vs. ALNS 40 ms	U	−0.25 [−4.02, +3.57]	not supported
3	H5	nominal fallback vs. no fallback	U	+1.97 [+1.61, +2.33]	supported
	H6	nominal fallback vs. buffer, high, windows ×0.5	U	+3.31 [+2.58, +4.03]	supported
	H7	nominal fallback vs. ALNS 40 ms	U	+4.61 [+4.15, +5.06]	supported
	H8	nominal fallback vs. constant buffer	U	+3.59 [+3.17, +4.02]	supported
4	H9	proposed vs. proportional buffer	U	−0.38 [−0.72, −0.05]	not supported
	H10	proposed vs. moment propagation (NI)	U	−0.25 [−0.62, +0.12]	NI not shown
	H11	nominal fallback within moment propagation	U	+9.66 [+9.04, +10.29]	supported
	H12	proposed vs. constant buffer	U	+4.22 [+3.74, +4.68]	supported
	H13	proposed vs. nominal model	U	+5.07 [+4.63, +5.52]	supported
4C	H14	tier C: proposed vs. nominal model	U	−0.57 [−3.49, +2.29]	not supported
	H15	tier C: nominal fallback vs. no fallback	U	+10.35 [+6.41, +14.39]	supported
5	H16	cost-aware vs. nominal fallback	u_1	+2.42 [+1.81, +3.02]	supported
	H17	cost-aware vs. proportional buffer tuned for λ	u_1	+0.90 [+0.38, +1.44]	supported, sup.
	H18	cost-aware vs. no fallback, z tuned for λ	u_1	+1.78 [+1.25, +2.31]	supported, sup.
	H19	cost-aware vs. nominal model	u_1	+11.75 [+10.88, +12.57]	supported
6	H20	proposed vs. CBBA	u_1	+12.77 [+11.91, +13.63]	supported
	H21	CBBA on the proposed model vs. CBBA	u_1	+12.12 [+11.31, +12.93]	supported
	H22	moment prop.: cost-aware vs. nominal fallback	u_1	+2.71 [+2.13, +3.27]	supported
	H23	moment prop.: cost-aware vs. no fallback	u_1	+3.30 [+2.83, +3.79]	supported
7a	H24	fixed-cost vs. proportional rule	u_1	+0.24 [−0.20, +0.67]	not supported
	H25	fixed-cost rule vs. buffer tuned for u_1	u_1	+1.34 [+0.77, +1.90]	supported, sup.
	H26	proportional rule vs. buffer tuned for u_1^w	u_1^w	+0.20 [−0.40, +0.79]	NI shown
7b	H27	nominal fallback vs. nominal model, shifted errors	U	4.83 to 5.67	supported, 4 of 4
	H28	cost-aware vs. nominal fallback, shifted errors	u_1	3.60 to 3.87	supported, 4 of 4

of every method by +1.29 to +1.56 pp. Supplementary Sections S3 and S4 report these results in full, together with an exact-execution benchmark, tier C flights and a scalability study with 40 UAVs. Study 2 moved the error model from the decision to stop into the planning model.

B. STUDY 2: THE CHANCE-CONSTRAINED MODEL AGAINST BASELINES

Study 2 comprises 48 600 tier B runs, that is, 270 held-out missions in 12 configurations with 15 methods (Supplementary Table S6). The proposed replanner, still without fallback, reached the highest utility, 52.15%. Its advantage over the six baselines of H1 ranged from +1.35 pp over the constant buffer to +2.87 pp over ALNS with 40 ms (all Holm-adjusted $p < 0.001$), and over the conventional planner it was +4.44 [+3.82, +5.05] pp, or 9.3% (H2). The proposed replanner served 5.72 tasks per mission after their deadline, 49% fewer than the 11.22 of ALNS with 40 ms. On the nominal model, the stopping rules hardly differed, with utilities between 49.28% and 49.66% although their computation ranged from 11.2 to 58.1 ms per call (Supplementary Fig. S4). On the chance-constrained model, the stopping rule was equivalent to a fixed budget of 40 ms (H3, 90% interval +0.22 [+0.09, +0.35] pp inside ± 0.5 pp), but it still guarded against long budgets: at $\kappa_B = 200$, the full budget of 250 ms lost 23.42 tasks per mission to late plans.

The gain was not uniform. The proposed replanner was worse than the constant buffer in 3 of the 12 configurations, all of them under high disturbances with halved windows, where it served −2.01 [−2.80, −1.21] pp relative to the buffer. With short windows and large errors, the effective deadline of many tasks falls before the earliest possible arrival, so the planner gives up tasks that a less cautious plan still serves in time. This weakness motivated the fallback of Section IV-D.

C. STUDY 3: NOMINAL FALLBACK FOR GIVEN-UP TASKS

Study 3 comprises 16 200 tier B runs on 270 new held-out missions (Supplementary Table S9). The replanner of study 2 again lost to the constant buffer under high disturbances with halved windows, by −1.95 [−2.72, −1.20] pp, so the weakness is systematic. The fallback variant reached the highest utility, 54.40%, and all four primary comparisons were significant (Holm-adjusted $p < 0.001$). It served +1.97 [+1.61, +2.33] pp more than the replanner of study 2 (H5), +3.31 [+2.58, +4.03] pp more than the constant buffer in the weak regime (H6), and +4.61 [+4.15, +5.06] and +3.59 [+3.17,

+4.02] pp more than ALNS with 40 ms and the constant buffer over all configurations (H7, H8). The fallback has a price in punctuality. A fallback task is planned against its nominal deadline and is often served late, and the fallback variant served 8.33 tasks late per mission, 46% more than the replanner of study 2. The case in which a late arrival has a cost of its own is the subject of study 5.

D. STUDY 4: PLANNING MODELS AND THE GENERALITY OF THE FALLBACK

Study 4 comprises 35 640 tier B runs on 270 new held-out missions (Table 5). All planning models use the same search with 40 ms and the periodic trigger, so they differ only in the margin and the fallback. The study replicates the main result on a third set of held-out seeds. With the fallback, the proposed model served 4.22 pp more reward than the constant buffer (95% CI 3.74 to 4.68, H12) and 5.07 pp more than the nominal model (4.63 to 5.52, H13), that is, 8.5% and 10.4% more, and 13.6% more than the conventional planner. All three advantages held in each of the four regimes (Supplementary Fig. S9), and with repair only and no ALNS the proposed model still served 3.33 pp more than the nominal one. Across the three independent sets of held-out seeds, every advantage over the constant buffer, the nominal model and the conventional planner was significant, and each grew once the fallback was added (Table 4, Supplementary Fig. S8).

The three models whose margin grows with the lead time reached utilities within 0.38 pp of each other. The proportional buffer served 0.38 pp more than the proposed model (-0.72 to -0.05, Holm-adjusted $p = 0.050$), so H9 is not supported. Moment propagation served 0.25 pp more, and the interval of the proposed model minus moment propagation, -0.62 to 0.12, crossed the margin of -0.5 pp, so non-inferiority was not shown (H10). The intercept and the mean correction of the calibrated model therefore add no utility beyond a tuned proportional margin. The closed form was cheaper, with 1.13 more search iterations per millisecond than moment propagation (13.35 against 12.22). The fallback carried over to moment propagation. Without it, moment propagation served 9.66 pp less (9.04 to 10.29, H11) and, at 44.44%, the lowest utility of all models, below the nominal model. A hard chance constraint on every task, however it is evaluated, thus gives up tasks that are still worth attempting. Among the secondary comparisons, online calibration served 0.30 pp less than the model with known disturbance level, the learned stopping rule added 0.34 pp, and the error model ranked the served tasks by risk, with 76.9% served on time under high disturbances against a predicted 71.3% (Supplementary Fig. S10).

E. SOFTWARE-IN-THE-LOOP FLIGHTS

In study 2, 120 held-out flights with mid disturbances and the original windows did not confirm the advantage of the replanner without fallback (H4; -0.25 [-4.02, +3.57] pp to ALNS with 40 ms, Holm-adjusted $p = 0.942$; Supplementary Section S5.2). Tier B predicted the utility of a flight closely ($r = 0.86$), but also expected only +2.82 [-1.20, +6.62] pp on these missions, too little to be confirmed by about 118 missions per method. Study 4C was therefore placed in the regime where tier B predicted the largest effect. It flew 108 flights on 36 new held-out missions with the complete replanner, with and without fallback, and ALNS with 40 ms on the nominal model. Every flight ran within the valid range of the real-time factor, with a mean of 1.96, and the interim analysis did not cross the O'Brien–Fleming bound.

The fallback carried over to the flights. With it, the replanner served 10.35 pp more reward than without it (95% CI 6.41 to 14.39, Holm-adjusted $p < 0.001$, H15), that is, 28.99% against 18.64%, or 56% more, and it was better in 28 of the 36 missions. Against the nominal model, it served 0.57 pp less (-3.49 to 2.29, $p = 0.699$), so H14 is not supported. Tier B predicted both outcomes (Fig. 5). On the same missions, its counterpart gave 0.22 pp less than the nominal model and 8.89 pp more than the model without fallback, with a correlation of $r = 0.83$ over 108 flights. The advantage over the nominal model that sized the study, 5.52 pp on development seeds, did not appear on the held-out missions even in tier B, so its absence in tier C reflects the sampled missions rather than the transfer to the autopilot. Over the full tier B grid, the advantage in this regime was 3.56 pp, which would need about 51 missions per method to be confirmed in tier C.

Fig. 4 illustrates the mechanism on one development mission, flown once with the proposed model with fallback and once with the nominal model under high disturbances with halved windows. The nominal model sent a UAV to a task that it reached after the deadline, while the proposed model served the same task on time with another UAV and then one more task, so it served 7 of the 15 tasks on time against 6, with 0 instead of 1 late arrival. These flights ran with the graphical interface, at a real-time factor of 1.22 to 1.34, and serve as an illustration only. All statistics in this section rest on the held-out flights without interface.

F. STUDY 5: THE COST OF A LATE ARRIVAL

The utility of studies 1–4 counts only tasks served on time, so a late service earns nothing and costs nothing. Scored again with a cost of λ mean task rewards for every late arrival, in an exploratory analysis without new runs (Supplementary Fig. S11), the advantage of the proposed model over the nominal model grew from 5.07 pp at $\lambda = 0$ to 14.19 pp at $\lambda = 2$. The fallback behaved in the opposite way. Its advantage vanished at a late cost λ^* of 0.5 in study 3, 1.1 for moment propagation in study 4 and 1.5 in tier C. The nominal fallback thus pays off only if a late arrival costs less than about one task reward. Study 5 tests the cost-aware fallback of Section IV-E on 270 new held-out missions (55 080 runs, Table 6, Fig. 6). For $\lambda = 0$, it is the nominal fallback by construction.

All four primary hypotheses were supported. At $\lambda = 1$, the cost-aware fallback reached a score of 45.81% against 43.40% for the nominal fallback, 2.42 pp more (95% CI 1.81 to 3.02, Holm-adjusted $p < 0.001$, H16), and served 33%

TABLE 5. Study 4, tier B held-out results: utility (%) per planning model overall and per regime (disturbance level and time-window factor, mean over κ_B), tasks served late per mission and search iterations per planning millisecond. All models use the same search with a fixed budget of 40 ms and periodic replanning. 270 missions, 12 configurations.

		Mid		High			
Planning model	All	×1	×0.5	×1	×0.5	Late	it/ms
Proposed planning model	53.85	73.07	48.05	57.01	37.27	7.54	13.35
Proposed, online calibration	53.55	73.22	48.25	55.26	37.48	7.63	13.59
Moment propagation	54.10	73.06	48.64	57.24	37.48	7.36	12.22
Moment propagation, no fallback	44.44	70.89	38.98	48.71	19.17	1.62	15.97
Proportional buffer $c\,h$	54.23	73.60	48.25	57.17	37.91	7.13	13.72
Constant 60 s buffer	49.63	69.46	43.72	51.68	33.66	6.52	14.75
Nominal model	48.78	66.73	44.89	49.79	33.71	11.22	13.00
Conventional (event-only)	47.40	65.13	43.54	48.48	32.46	10.28	9.04

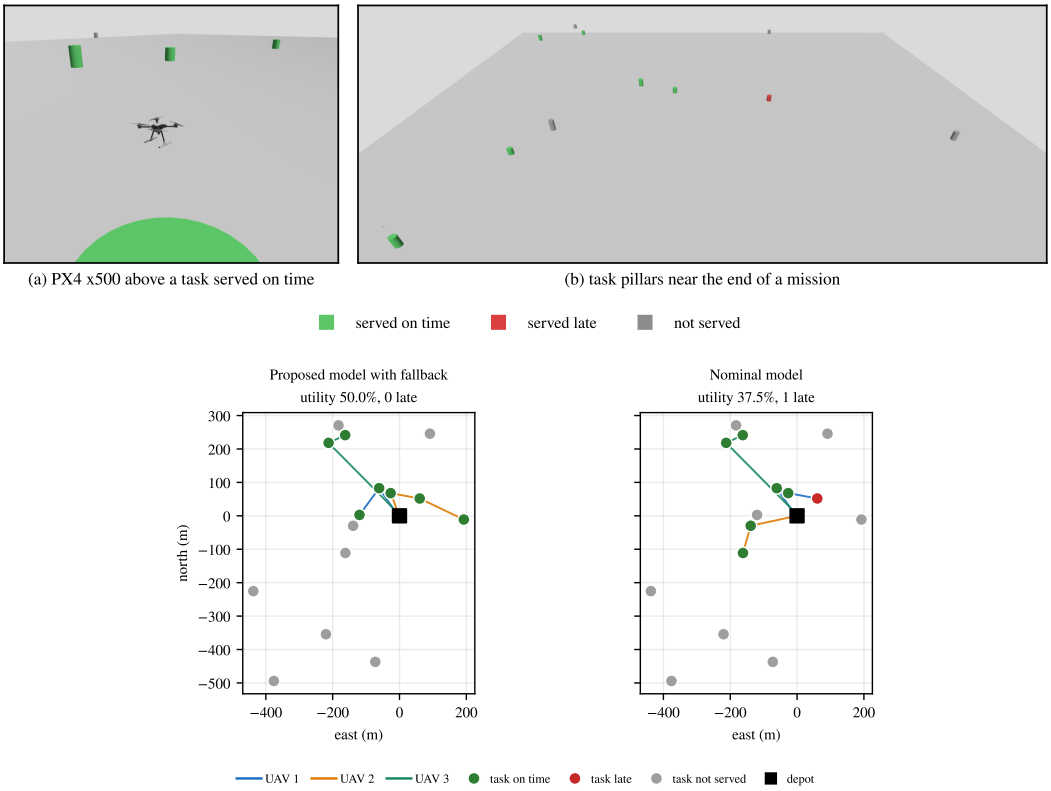


FIGURE 4. Illustrative tier C flight on a development mission (seed 1, high disturbances, windows ×0.5). Top: Gazebo scene with PX4-controlled X500 quadrotors; every task is a pillar coloured by its outcome. Bottom: top view of the trajectories recorded by PX4 for the proposed model with fallback and the nominal model on the same mission. These flights ran with the graphical interface and are not part of any statistical test.

fewer tasks late. Against the proportional buffer tuned for $\lambda = 1$, non-inferiority was shown, and in the fixed sequence it was better by 0.90 pp (0.38 to 1.44, $p < 0.001$, H17). Against the model without fallback with z tuned for $\lambda = 1$, it was better by 1.78 pp (1.25 to 2.31, $p < 0.001$, H18), and over the nominal model by 11.75 pp (H19). Both tuned competitors had been selected on development missions from sixteen and five candidates for the late cost of the score, whereas the cost-aware fallback used the z of study 4. The advantage came from high disturbances, where it served 2.10 and 3.49 pp more than the two tuned competitors. Under mid disturbances, the methods were on par, and with the original windows both tuned competitors were better, by 2.08 and 1.68 pp.

In practical units, per 100 tasks the cost-aware fallback served 50.2 on time and 7.3 late, the nominal fallback 51.3 and 10.9, and the nominal model 48.3 and 15.6. A late arrival was thus 13.8% of all arrivals, against 18.9% and 25.7%, and the fleet flew 92.2 km per mission instead of 101.5 and 104.5 km, consistent with fewer flights to tasks that were likely to be reached too late.

At other late costs, the picture was mixed (Fig. 6). At $\lambda = 0.5$, the cost-aware fallback was on par with the tuned buffer (-0.22 to 0.64 pp) and better than the tuned model

TABLE 6. Study 5, tier B held-out results: mean score u_λ (%) for a late arrival that costs λ mean task rewards ($\lambda = 0$ is the utility of studies 2–4), and tasks served late per mission at $\lambda = 1$. The cost-aware fallback and the tuned competitors use the late cost of the column. 270 missions, 12 configurations each.

Method	$\lambda = 0$	0.5	1	2	Late
Cost-aware fallback (proposed)	54.33	49.49	45.81	39.71	4.96
Nominal fallback (study 4)	54.33	48.86	43.40	32.46	7.39
Proportional buffer, tuned per λ	–	49.28	44.91	40.73	4.36
No fallback, z tuned per λ	–	47.84	44.03	39.73	3.00
Moment propagation, fallback	54.57	49.14	43.72	32.87	7.29
Moment propagation, no fallback	45.06	44.00	42.94	40.81	1.50
Constant 60 s buffer	50.33	45.66	40.98	31.64	6.49
Nominal model	49.63	41.85	34.07	18.50	11.08
Conventional (event-only)	47.84	40.56	33.27	18.71	10.25

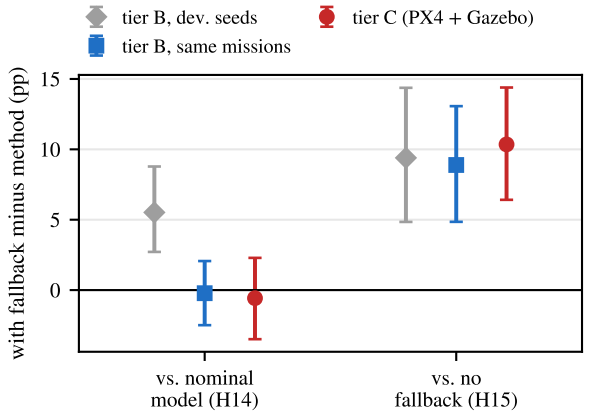


FIGURE 5. Study 4C, held-out seeds 9000–9011, high disturbances with halved windows: utility of the proposed replanner with fallback minus ALNS with 40 ms on the nominal model and minus the replanner without fallback, with 95% paired bootstrap intervals. Tier C (36 missions, PX4 and Gazebo) is shown with its tier B counterpart on the same missions and with the tier B estimate on development seeds from which the size of the study was computed.

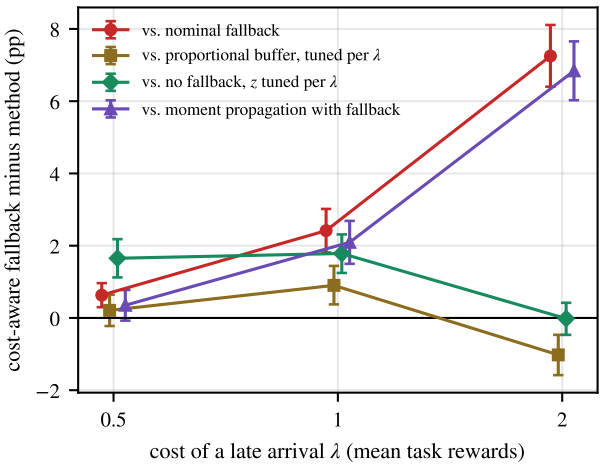


FIGURE 6. Study 5, tier B, held-out seeds 10000–10029: score u_λ of the cost-aware fallback minus each method, when a late arrival costs λ mean task rewards. The cost-aware fallback and the tuned competitors use the λ of the score. 95% paired bootstrap intervals over 270 missions.

without fallback and the nominal fallback. At $\lambda = 2$, it stayed 7.25 pp above the nominal fallback and matched the tuned model without fallback, but the proportional buffer tuned for $\lambda = 2$ was better by 1.02 pp (-1.58 to -0.47), and moment propagation without fallback by 1.10 pp. At this cost, a fallback task needs a probability of at least 0.67 of meeting its deadline, and the rule ignores the delay that it causes to the tasks after it on the same route, which weighs more when lateness is expensive. The competitors, however, have to be tuned again for every late cost. Tuned for $\lambda = 1$, the proportional buffer stayed 1.05 pp behind the cost-aware fallback at $\lambda = 2$ and 1.46 pp behind at $\lambda = 0.5$. A wrong late cost also costs the cost-aware fallback, 1.22 pp when it was run with $\lambda = 1$ and scored at $\lambda = 2$. When a late arrival costs nothing, the rule run with $\lambda = 1$ served 1.20 pp less utility than the nominal fallback.

G. STUDY 6: A PUBLISHED ALLOCATOR AND THE GENERALITY OF THE COST-AWARE FALLBACK

Study 6 comprises 29 160 tier B runs on 270 new held-out missions (Table 7, Fig. 7). Every method with the cost-aware fallback was run with $\lambda = 1$, and the scores for other late costs are computed from the same runs. All four primary hypotheses were supported, each with a Holm-adjusted $p < 0.001$. With the score u_1, the proposed replanner exceeded CBBA by 12.77 pp (95% CI 11.91 to 13.63, H20), and CBBA itself gained 12.12 pp (11.31 to 12.93, H21) when it planned on the deadlines of the cost-aware fallback. Within moment propagation, the cost-aware fallback scored 2.71 pp above the nominal fallback (2.13 to 3.27, H22) and 3.30 pp above the model without fallback (2.83 to 3.79, H23). The advantage of the cost-aware over the nominal fallback in the proposed model, 2.42 pp (1.91 to 2.92), replicates H16 on a second set of seeds.

The difference to CBBA lies in the planning model and not in the search or in the cost of computation. CBBA converged after 2.8 consensus rounds and 5.1 ms per call on average, against 40 ms for ALNS, and on the nominal model it reached

TABLE 7. Study 6, tier B held-out results: mean score u_λ (%) for a late arrival that costs λ mean task rewards ($\lambda = 0$ is the utility), tasks served late per mission and planning time per call. Methods with the cost-aware fallback use $\lambda = 1$. 270 missions, 12 configurations each.

Method	$\lambda = 0$	0.5	1	2	Late	ms
Proposed (ALNS, cost-aware fallback)	53.42	49.81	46.21	39.01	4.91	40.2
CBBA on the proposed model	51.82	48.69	45.56	39.30	4.25	2.1
CBBA, periodic	48.56	41.00	33.44	18.32	10.57	5.1
CBBA, event-triggered	46.69	39.58	32.46	18.23	9.77	13.2
Moment propagation, cost-aware fallback	52.01	49.49	46.98	41.95	3.52	40.3
Moment propagation, nominal fallback	54.97	49.62	44.27	33.57	7.18	40.2
Moment propagation, no fallback	45.75	44.71	43.67	41.60	1.51	40.2
Nominal model (ALNS)	49.55	41.76	33.98	18.42	11.02	40.1

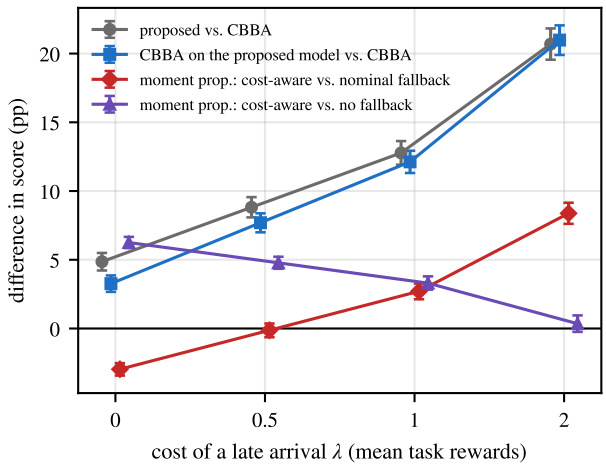


FIGURE 7. Study 6, tier B, held-out seeds 12000–12029: differences in the score u_λ between methods, with 95% paired bootstrap intervals over 270 missions. All methods with the cost-aware fallback were run with $\lambda = 1$ and are scored at every λ from the same runs.

almost the utility of ALNS (-1.48 to -0.51 pp). On utility alone ($\lambda = 0$), the proposed replanner served 4.86 pp, or 10.0%, more than CBBA, and 6.72 pp more than CBBA with the event trigger. CBBA served 10.57 tasks late per mission, the proposed replanner 4.91. On the proposed model, CBBA came within 0.65 pp of the proposed replanner at $\lambda = 1$, with 2.1 ms per call, so most of the gain does not depend on ALNS. The exception was the regime of high disturbances with halved windows at $\lambda = 0$, where the proposed replanner and CBBA were on par (-0.24 [-1.29, +0.82] pp), because a fallback for $\lambda = 1$ gives up tasks that a late arrival would not penalise at $\lambda = 0$.

Within moment propagation, the cost-aware fallback behaved as in the closed form. Without a late cost it lost 2.96 pp against the nominal fallback, it broke even at $\lambda = 0.5$, and its advantage grew to 8.38 pp at $\lambda = 2$. With the cost-aware fallback, moment propagation scored 0.76 pp more than the closed-form model at $\lambda = 1$ (closed form minus moment propagation -1.31 to -0.24) and 2.94 pp more at $\lambda = 2$, whereas the closed form was 1.41 pp better at $\lambda = 0$. Once lateness has a cost, the variance that moment propagation tracks leg by leg therefore pays off, even though it ran fewer search iterations per call (433.2 against 467.4).

H. STUDY 7: LATE-COST MODEL AND ERRORS OUTSIDE THE MODEL

Study 7 comprises 55 080 tier B runs in study 7a and 56 700 in study 7b, each on 270 new held-out missions (Fig. 8, Supplementary Section S11). For the scores u_λ and u^w_λ, the development selection of the tuned buffer differed only at $\lambda = 1$ under high disturbances, with $c = 0.45$ for u_1 and $c = 0.6$ for u^w_1, both without fallback.

Matching the rule to the score changed little at a late cost of one task reward. With u_1, which charges a fixed cost per late arrival, the fixed-cost rule scored 45.11% and the proportional rule of studies 5 and 6 44.87%, a difference of +0.24 [-0.20, +0.67] pp (Holm-adjusted $p = 0.286$), so H24 is not supported. Against the proportional buffer tuned for u_1, at 43.78%, the fixed-cost rule was non-inferior and better by +1.34 [+0.77, +1.90] pp ($p < 0.001$, H25), which replicates H17 on a new set of seeds. With u^w_1, which charges a cost proportional to the reward, the proportional rule was non-inferior to the buffer tuned for that score (+0.20 [-0.40, +0.79] pp, Holm-adjusted $p = 0.022$, H26), but not better ($p = 0.499$). With u'_1, which charges exactly the fixed cost $\bar{w}$ that the rule assumes instead of the realised mean reward of the mission, the two differences were +0.26 [-0.17, +0.69] and +1.35 [+0.79, +1.91] pp (Supplementary Table S12), so the difference between the two definitions of the fixed cost does not affect these conclusions.

The match mattered more at a higher cost, and each rule did better on its own score. The fixed-cost rule scored +0.85 [+0.32, +1.39] pp above the proportional rule with u_2, and the proportional rule scored 0.86 and 1.25 pp above the fixed-cost rule with u^w_1 and u^w_2. With u_2, the matched rule showed no significant difference to the buffer tuned for this score (+0.10 [-0.63, +0.82] pp), whereas the proportional rule stayed significantly behind it, as in study 5 (-0.75 [-1.38, -0.12] pp). The interval of the matched rule does not establish equivalence either, since it extends beyond the margin of ± 0.5 pp. With u^w_2, both rules stayed behind the buffer tuned for that score, the matched proportional rule by 1.36 pp. At $\lambda = 0.5$, all point estimates were within half a point, with no significant difference to the tuned buffer. The mismatch in studies 5 and 6 thus lowered the score of the cost-aware fallback slightly

rather than raised it, and the weakness at twice the cost remains for a late cost that is proportional to the reward.

Errors that the model does not describe lowered the utility of every method only slightly. The nominal model served 42.28% under the calibrated errors and between 40.09% with rare long delays and 42.67% with Student-t errors, so the departures were moderate. The advantages of the proposed model held under all four of them, and all eight primary comparisons were significant (Holm-adjusted $p < 0.001$). With the nominal fallback, the proposed model served 4.83 to 5.67 pp more than the nominal model (H27), against 5.34 pp under the calibrated errors. The smallest advantage occurred with a speed bias per UAV, which correlates the errors along a route in a way the model does not represent. On u_1, the cost-aware fallback scored 3.60 to 3.87 pp more than the nominal fallback (H28), and 2.94 to 3.64 pp more than the proportional buffer of study 4. Rare long delays raised the mean finish-time error of the proposed model from 78.8 to 88.3 s, and its predicted probability became slightly optimistic in the upper tail, with 92.6% of the tasks served on time against a predicted 93.6% (Supplementary Section S9). Under the other departures, the upper tail stayed within about one point of its prediction, on the conservative side for Student-t errors.

VII. DISCUSSION AND LIMITATIONS

The seven studies answer the four research questions as follows. On RQ1, the spread of the finish-time error grows almost linearly with the lead time, and a chance constraint built on it reduces to one effective deadline per task that fast heuristics use unchanged. On RQ2, the planning model matters most, the trigger second and the stopping rule least. On RQ3, the replanner served more reward than a constant buffer, the nominal model, a conventional planner and CBBA, and as much as a tuned proportional buffer and moment propagation, also under the tested departures from its error model, but software-in-the-loop flights confirmed only the fallback. On RQ4, the probability of meeting the deadline decided which given-up tasks to attempt without tuning. With a fixed late cost of one task reward, this beat competitors tuned for that cost on two sets of seeds, and with a cost proportional to the reward it was as good as them, but not at twice that cost when the cost is proportional to the reward.

The order of priorities holds across the designs. A margin that grows with the lead time raised the utility by more than any choice of stopping rule or trigger, on three independent sets of held-out seeds, without search and with CBBA in place of ALNS. Periodic replanning helped every method on the nominal model, but only by 0.22 pp on the chance-constrained model, which suggests that both correct the same accumulated deviations, one before and one after they occur. The stopping rule was equivalent to a matched fixed budget in study 2 and added 0.34 pp in study 4. Its main value lies in avoiding the losses of long budgets on slow planners, probably because the improvement curve of this ALNS flattens within a few tens of milliseconds.

The shape of the margin matters more than the way it is computed. The constant buffer was clearly worse than every margin that grows with the lead time, whereas the calibrated effective deadline, a tuned proportional buffer and moment propagation were within 0.38 pp of each other in utility. Because the spread grows almost linearly, a margin $c\,h$ with a suitable slope captures most of it, and under high disturbances the tuned slope nearly equalled the one that the calibrated model implies. Moment propagation evaluates the actual legs of a route and reduced the search rate, but with a cost of lateness its detail paid off in study 6, by 0.76 pp at one task reward and 2.94 pp at two. The closed form ran 1.13 more search iterations per millisecond in study 4, a modest gain, and the studies did not vary the planning budget, so they do not show that it wins under tighter budgets. Its advantage lies elsewhere. It needs one division per task, no model of the individual legs and no change to the feasibility checks of a heuristic, which is what allowed CBBA to use it unchanged. Where lateness is costly and the planner can afford to evaluate every leg, the more detailed model scored higher. The comparison with CBBA places the gain in the planning model rather than in the allocation method. On the proposed model, CBBA came within 0.65 pp of the proposed replanner while it planned for 2.1 ms per call, so the effective deadlines can be handed to an auction-based allocator without changing its bidding rule.

The probability of meeting the deadline is where the calibrated model earns its place. A hard chance constraint discards every task whose probability falls below $\Phi(z)$, even when a vehicle has nothing better to do, and study 4 shows that this holds however the constraint is evaluated. The nominal fallback recovers these tasks but serves many of them late. With utility alone, a tuned proportional buffer was as good as the calibrated model. Once a late arrival has a cost, the question is no longer only how large the margin is but which tasks are worth attempting at all, and that requires a probability. The closed form turns the break-even rule of Eq. (6) into another effective deadline, which served more than a buffer and a safety factor tuned for one task reward and stayed ahead of the buffer whenever the buffer had been tuned for a different cost. Applied to moment propagation, the same rule behaved in the same way, so it is not tied to the closed form. The rule depends on the cost model, and study 7a shows that this matters less than expected. Run with the cost model of the score, it did not differ significantly from the mismatched rule at one task reward. At twice the cost, only with a fixed cost did the difference to the tuned buffer cease to be significant, without establishing equivalence. When the cost was proportional to the reward, a tuned buffer without fallback stayed better at twice the cost. The rule prices each attempt on its own and ignores the delay that a fallback task causes to later tasks on its route, which weighs more when lateness is expensive, and a route-level version that also charges this delay is the natural next step.

The guarantee of the effective deadline rests on a Gaussian error model, but the results did not depend on it under the tested model departures, which does not imply robustness

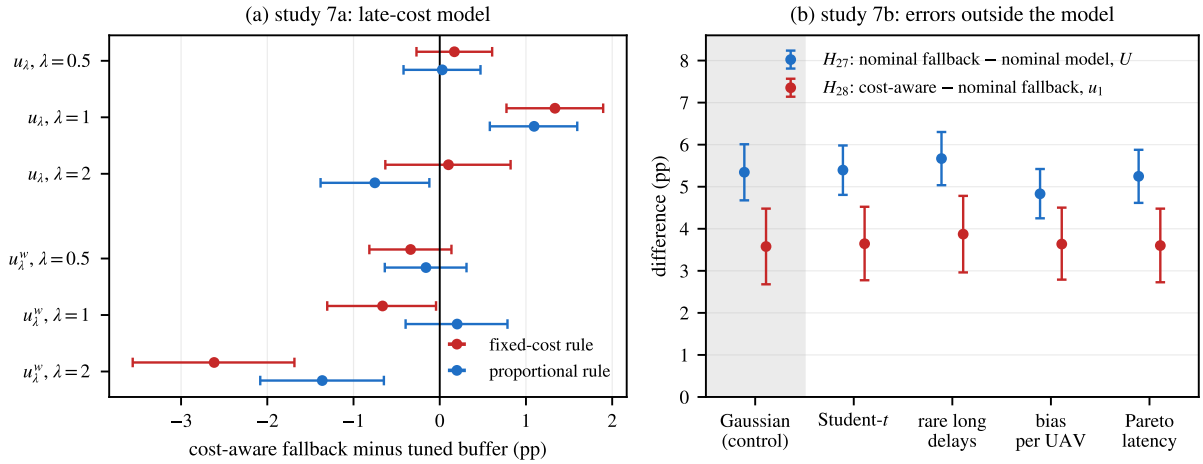


FIGURE 8. Study 7, tier B. (a) Study 7a, held-out seeds 13000–13029: cost-aware fallback with the fixed-cost and with the proportional rule minus the proportional buffer tuned for the same score and late cost. (b) Study 7b, held-out seeds 14000–14029, high disturbances: the two primary comparisons under the calibrated Gaussian errors (shaded control, not part of the primary family) and under the four tested departures from the error model, namely Student-t speed and service errors with three degrees of freedom, rare long delays (probability 0.1 per leg, exponential with a mean of 90 s), a persistent speed bias per UAV (standard deviation 8%) and a Pareto dissemination delay (tail index 1.5). 95% paired bootstrap intervals over 270 missions.

to arbitrary execution errors. Heavy-tailed errors, rare long delays, a speed bias per UAV and heavy-tailed latency left the advantage over the nominal model between 4.83 and 5.67 pp, against 5.34 pp under the calibrated errors. The bias per UAV, which correlates the errors along a route, reduced it most. The model ranks tasks by their risk, and a ranking is less sensitive to the shape of the tails than the probability itself, which became slightly optimistic in the upper tail only with rare long delays.

The two tiers play different roles. The kinematic tier resolves differences of a few percentage points over tens of thousands of missions, whereas a flight with autopilot and vehicle dynamics is costly. Tier B predicted every tier C comparison on the same missions, so the flights support the transfer of the kinematic results and confirm the fallback, but the advantage over the nominal model rests on tier B, and tier C studies should be sized from a conservative effect.

Several limitations bound these conclusions. The error model is linear in the lead time and fitted per disturbance level, so the level is assumed to be known. Online calibration removes this assumption but served 0.30 pp less reward, and its non-inferiority was not shown. The guarantees of Propositions 1 and 2 hold under the Gaussian error model, and study 7b covered only four departures from it, at sizes fixed in advance and under high disturbances. The chance constraint is imposed on every task separately rather than jointly on a route, and the cost-aware fallback is a local break-even rule. The fallback, the models of study 4, the cost-aware fallback and study 7 were designed after the held-out results of earlier studies and were therefore tested on new seeds. The cost-aware fallback assumes that the cost of a late arrival is known. Study 7a covered a cost proportional to the reward and a fixed cost, but not a cost that grows with the lateness, and the rule was tested in tier B only. CBBA ran centrally with a fully connected network, the most favourable case for its communication, so its robustness to lost links was not part of the comparison. Tier C uses three UAVs and fifteen tasks per mission, and software-in-the-loop simulation reproduces the autopilot and the vehicle dynamics but not the sensors, the radio link or the weather of a field test. No field flights were made, and the rescaled tier C missions share their disturbance model with tier B, so the transfer to real vehicles remains to be shown. The latency factor emulates a slower planning computer instead of measuring one, and because the stopping rules read a wall clock, the mean utility of a method varied by up to 0.12 pp between two runs of the same missions. Finally, all methods share one family of scenarios, and other mission types may change the size of the gains.

VIII. CONCLUSION

This paper proposed LTCC, a multi-UAV replanner that plans against a chance constraint on the finish time of every task. Because disturbances persist over a leg, the spread of the finish-time error grows linearly with the lead time, and with an affine error model the chance constraint reduces exactly to an effective deadline per task, so fast heuristics are reused unchanged. In seven studies on held-out missions with protocols frozen in advance, the margin that grows with the lead time, together with the fallback, carried the gain in utility. The replanner served more than the nominal model, a constant buffer, a conventional planner and, by 10.0%, CBBA, and as much as a tuned proportional buffer and moment propagation, and its advantage over the nominal model persisted under four tested departures from its error model. The probability of meeting a deadline, which a buffer lacks, lets the replanner price lateness. With a late arrival that costs one task reward, the cost-aware fallback served 2.42 percentage points more than the nominal fallback and more than a buffer tuned for

this cost, without tuning of its own, on two sets of seeds. For a late cost proportional to the reward, it was as good as a buffer tuned for that cost, and it improved moment propagation in the same way. At twice this cost, a tuned buffer was better when the cost was proportional to the reward. Software-in-the-loop flights with PX4 and Gazebo confirmed the fallback and agreed closely with the kinematic tier, but did not resolve the advantage over the nominal model. Future work will charge the delay that a fallback task causes on its route, fly the cost-aware fallback in the software-in-the-loop tier, run it inside a distributed allocator, and test the replanner on embedded hardware and in field flights.

DATA AND CODE AVAILABILITY

The planner, both simulation tiers, the experiment configurations with their SHA-256 hashes, the written protocols with the commit of every freeze and every deviation, all raw results and the scripts that produce every number, table and figure of this paper are available from the authors upon reasonable request.

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